

JULIAN ZHANG

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EDUCATION

University of Waterloo (Sep 2024 - Apr 2029)
Computer Science and Finance Double Major (CFM) | *President's Scholarship of Distinction*

University of Toronto Schools (Sep 2018 - Jun 2024)
Ontario Secondary Schools Diploma | *Cumulative GPA: 4.0*

EXPERIENCE

Oakville Centre For Vision - Software Engineer (May 2024 - Sep 2024)

- Created a web platform for eye-tracking goggles used to analyze vision patterns in professional athletes, collaborating directly with optometrist Dr. Ronald Gall
- Focused on data analysis of tests that analyzed gaze vectors for saccades, fixations, and pursuits
- Collaborated with a hardware team to modernize a **6000+ line** codebase for improved functionality

University of Toronto Clean Energy Lab - Research Assistant (Jun 2022 - Sep 2023)

- Applied Crystal Graph Convolutional Neural Network (CGCNN) to examine trends in chemical properties
- Analyzed results for material discovery with a focus on advancing lithium-ion alternatives for sustainable energy. Synthesized findings for practical applications of mitigating climate change
- Presented findings to Prof. Oleksandr Voznyy and PhD Students as the group's youngest member

Canadian National Debate Team - Member (May 2023 - Aug 2024)

- Represented Canada as one of 12 debaters at the 2024 Belgrade World School Debating Championship (WSDC) 2024, honing advanced critical analysis, structured thinking, and public speaking skills
- Conducted research across politics, economics, and international relations, engaging with field experts
- Organized 5+ debate tournaments attracting **500+** participants and raising **\$10,000+** for charity

PROJECTS

- SignWave (JamHacks Winner):** AI translator converting text and audio into sign language animations, improving accessibility for the deaf and hard of hearing (OpenCV, Flask, MediaPipe, Whisper)
- Common Ground:** Interactive social media graph visualization platform for Twitter/X, facilitating connection discovery through an LLM-based recommendation engine (D3.js, React.js, Vite, Groq)
- zhaju.github.io:** Personal website featuring a blog and repository of high school and university course notes, supporting **50+ students** with educational content (HTML/CSS, TypeScript, Node.js, Quartz)

SKILLS

- Languages:** C/C++/C#, Python, Racket, Java, JavaScript, HTML/CSS, LaTeX
- Libraries:** React.js, Node.js, NumPy, matplotlib, pandas, PyTorch, TensorFlow, Yahoo Finance
- Developer Tools:** Git, GitHub, Conda, VSCode, PyCharm, CLion, Jupyter Notebook
- Certifications:** Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning

SELECTED AWARDS

- JamHacks VII Hackathon - **First Prize Overall, Best Use of GitHub**
- American Invitational Math Exam (AIME) Qualifier - **Top 8% Worldwide**
- Canadian National British Parliamentary Debate Championship - **1st speaker & team**
- Munk Debates ReSolved Global Debate Competition - **Champion, awarded \$5,000**